MSA Case Studies - Real-World GRR Failures and Fixes

4 case studies: what went wrong, root cause, lesson learned

QUALITY HEAD AGENTS | AIAG MSA 4th Edition | IATF 16949:2016

CS-MSA01 - CNC Bore Diameter: GRR 42% - Auditor NC

Product:	Engine block bore diameter, CC characteristic
Problem:	IATF auditor asked for GRR report for bore diameter gauge (air gauge). GRR was 42% - 'Unacceptable'. Major NC raised.
Root Cause:	Air gauge had worn out nozzles. Calibrated correctly but nozzle condition created excessive variation in repeatability.
Lesson:	Calibration does not equal GRR acceptance. A calibrated gauge can still have unacceptable GRR if it is worn or damaged.
Corrective Action:	Replaced air gauge nozzles. Repeated GRR study - result: 7.2%. Updated MSA plan with 6-monthly GRR re-study.
Best Practice:	GRR must be re-studied after gauge maintenance, repair, or at defined intervals. Do not wait for the auditor.

CS-MSA02 - Hardness Tester: Linearity Study Failure

Product:	Heat-treated components, 58-62 HRC requirement
Problem:	Hardness tester passed GRR (study variation 12%). But on linearity study, R^2 = 0.71 - non-linear response.
Root Cause:	Tester was accurate at 58 HRC (middle of range) but showed increasing positive bias above 62 HRC. Soft indenter causing creep at high loads.
Lesson:	GRR alone is not sufficient for gauges used across a range. Linearity study is required.
Corrective Action:	Replaced Vickers indenter. Linearity re-study: R^2 = 0.97. Included linearity as annual requirement in gauge register.

CS-MSA03 - Visual Inspection: Attribute GRR Reveals Operator Disagreement

Product:	Surface finish inspection - customer visible parts
Problem:	Customer complaint: 'scratched parts supplied.' Internal inspection used visual assessment (3 inspectors, subjective criteria).
Root Cause:	Attribute GRR study showed Kappa = 0.41 between inspectors. Borderline parts were classified differently by each inspector.
Lesson:	Subjective visual inspection without defined criteria has unacceptable Kappa. Every accept/reject decision must have measurable criteria.
Corrective Action:	Implemented visual standard limit samples (golden samples). Attribute GRR re-study: Kappa = 0.82.

CS-MSA04 - Thread Gauge: Stability Drift After 3 Months

Product:	M12 x 1.5 threaded boss, Go/No-Go gauge inspection
Problem:	3 months after PPAP approval, customer receiving threads that Go gauge would not accept. Internal gauge was 'passing' these parts.
Root Cause:	Go gauge worn past tolerance due to high-volume inspection. Stability study had not been set up after PPAP.
Lesson:	GRR at PPAP is a snapshot. Ongoing stability monitoring is required to detect gauge wear and drift.
Corrective Action:	Replaced gauge. Implemented stability study: measure reference part at start/mid/end of each shift. Action limit defined at 50% of tolerance.